

Round 2 Evaluation Metrics Report

Dataset and evaluation design

The dataset contains 9,000 labelled text records. Two supervised NLP tasks were evaluated:

1. Sentiment classification.
2. Topic-category classification.

A group-aware split prevented identical text values from appearing in more than one data split.

Split	Records
Training	5382
Validation	1805
Testing	1813

Final metrics were calculated only on the untouched test set.

1. Sentiment classification

Target

```
post_text -> sentiment_label
```

The classes are Negative, Neutral, and Positive.

Model selection

Candidate model	Accuracy	Macro F1	Weighted F1
logistic_regression	0.5961	0.5972	0.5971
linear_svm	0.5801	0.5806	0.5806

Selected model: **logistic_regression**

Final test metrics

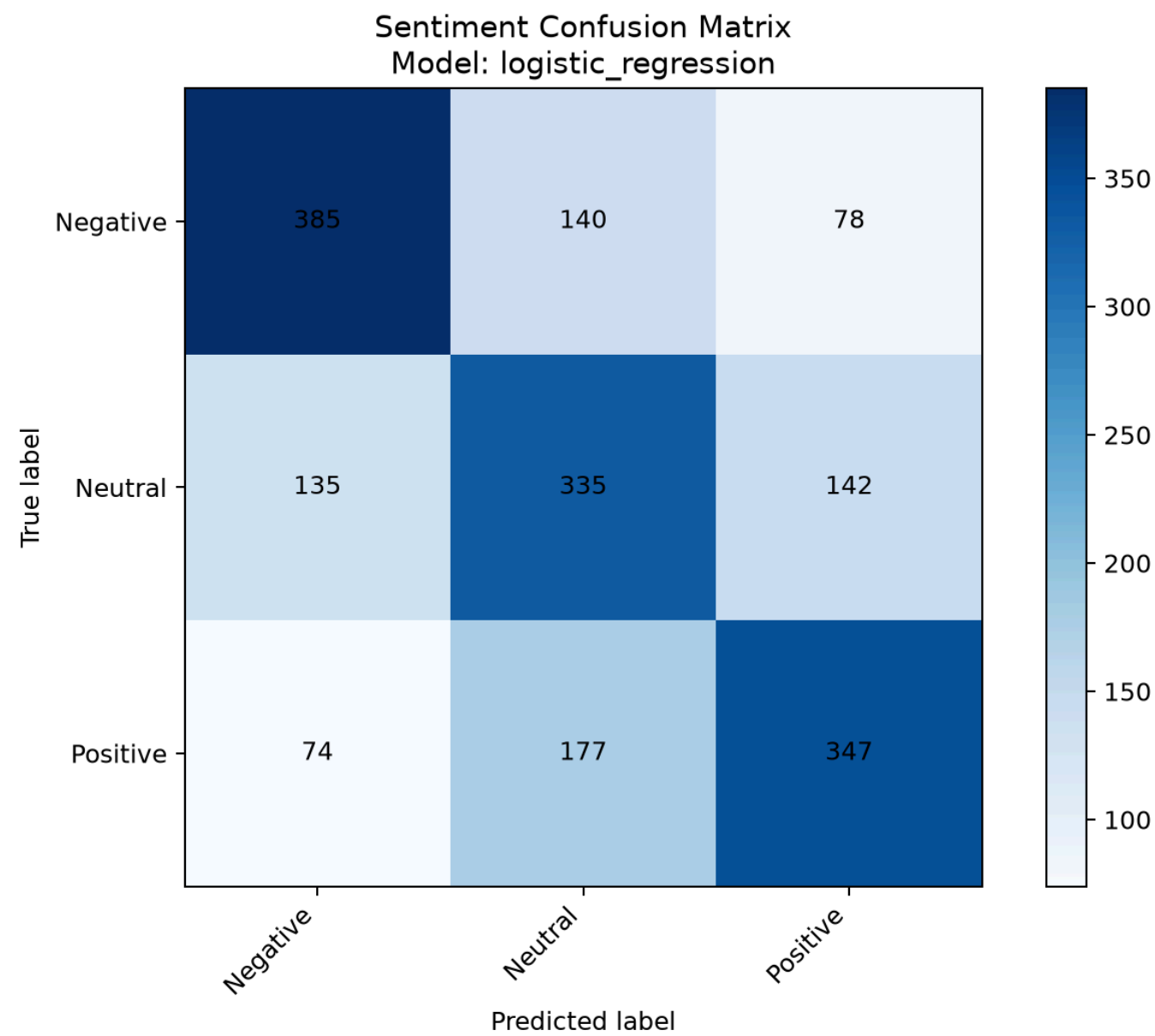
Metric	Score
Accuracy	0.5885
Macro precision	0.5913

Metric	Score
Macro recall	0.5887
Macro F1-score	0.5897
Weighted F1-score	0.5894

Per-class metrics

Class	Precision	Recall	F1-score	Support
Negative	0.6481	0.6385	0.6433	603
Neutral	0.5138	0.5474	0.5301	612
Positive	0.6120	0.5803	0.5957	598

Confusion matrix



2. Topic classification

Target

```
post_text -> topic_category
```

The topic classes are Account Security, Community Discussion, Feature Feedback, and Technical Issues.

Model selection

Candidate model	Accuracy	Macro F1	Weighted F1
logistic_regression	0.9114	0.6049	0.8961
linear_svm	0.9158	0.5884	0.8961

Selected model: **logistic_regression**

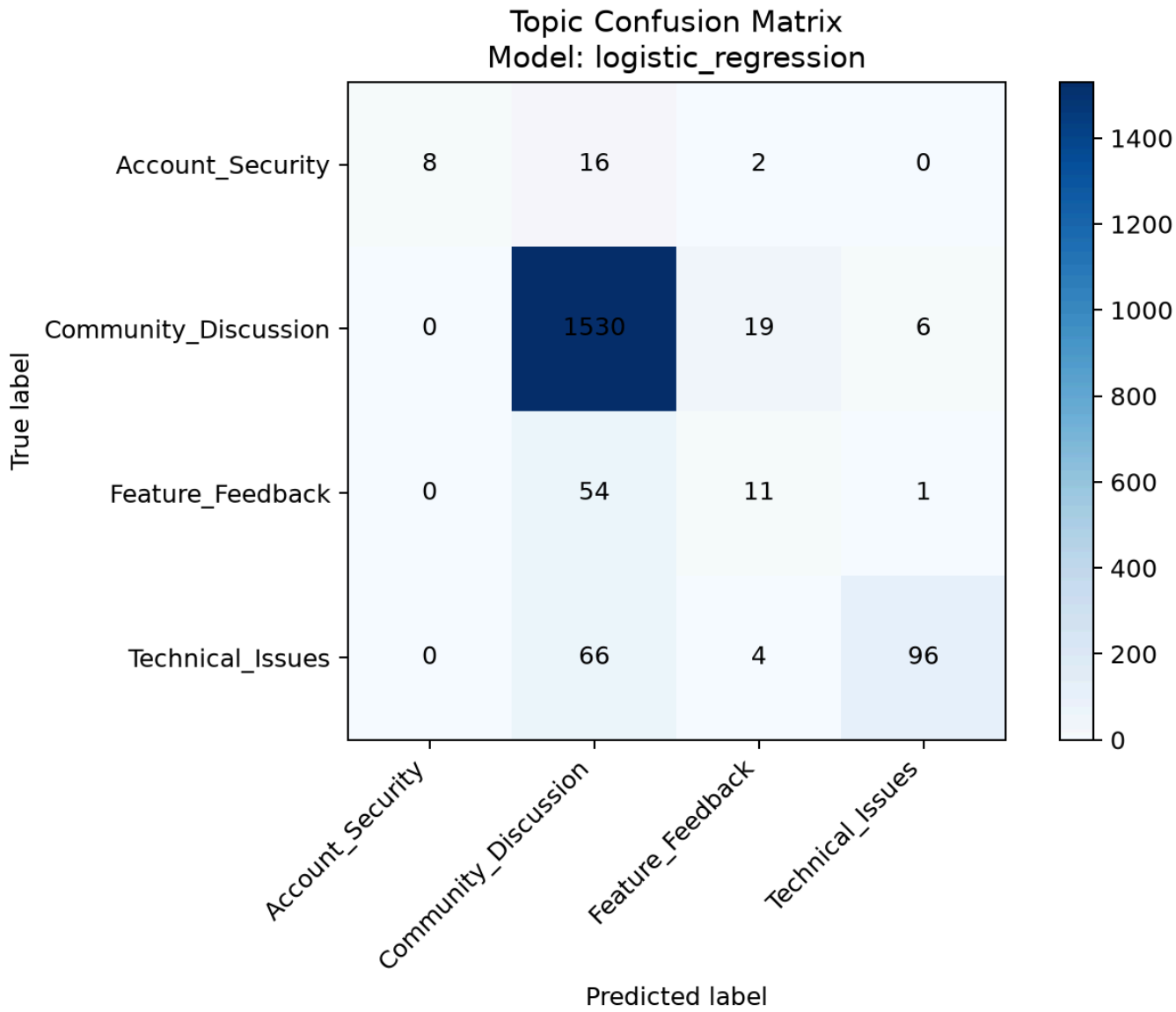
Final test metrics

Metric	Score
Accuracy	0.9073
Macro precision	0.7890
Macro recall	0.5091
Macro F1-score	0.5875
Weighted F1-score	0.8948

Per-class metrics

Class	Precision	Recall	F1-score	Support
Account_Security	1.0000	0.3077	0.4706	26
Community_Discussion	0.9184	0.9839	0.9500	1555
Feature_Feedback	0.3056	0.1667	0.2157	66
Technical_Issues	0.9320	0.5783	0.7138	166

Confusion matrix



3. Interpretation

The sentiment classifier achieved a macro F1-score of **0.5897**.

The topic classifier achieved accuracy of **0.9073** and macro F1-score of **0.5875**.

The topic accuracy is high partly because **Community_Discussion** is the dominant class. Macro F1 is therefore also reported.